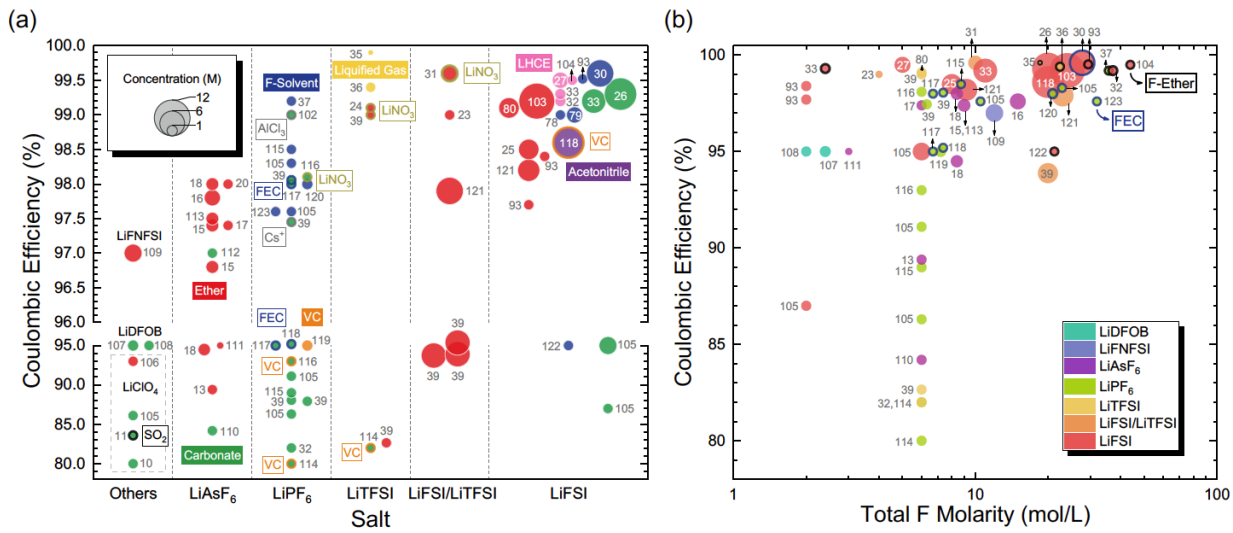

Figure 2 | Coulombic efficiency of select electrolyte systems. (a) Compendium of reported CE of published electrolytes categorized by salt species (horizontal axis), solvent (marker fill color), additive (marker border color), and salt concentration (marker size). (b) CE as a function of total atomic fluorine content in both the salt and solvent, estimated using salt concentration and solvent density. Cycling protocols and additional details (current collector, current density, cell construction) for a-b can be found in the **Source Data** file. References not cited elsewhere in the text are collected here.[105-124]

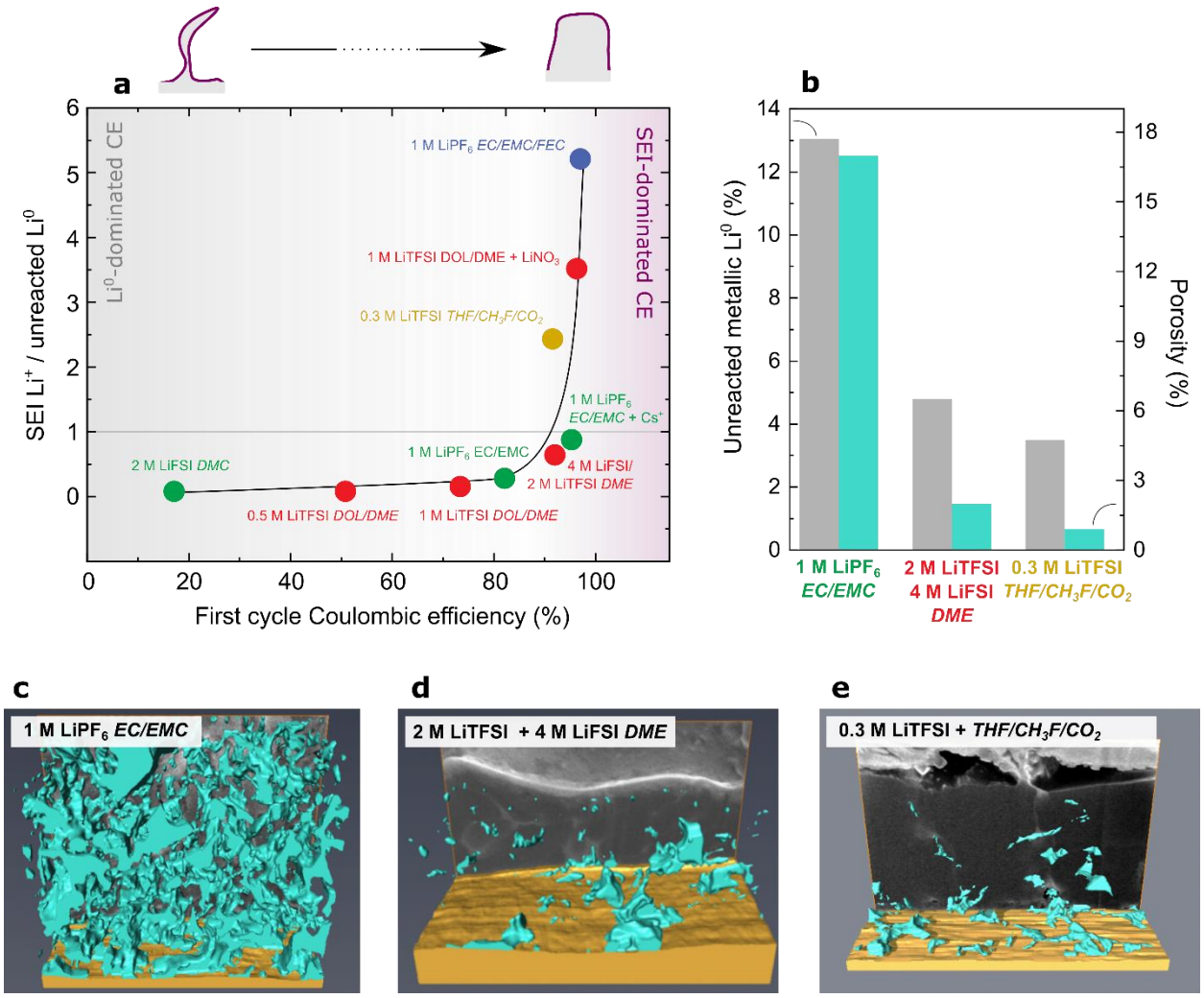

Figure 3 | Quantifying morphology and composition of electrochemically-inactive Li. (a) Ratio of capacity lost as SEI Li^+ to capacity lost as encapsulated unreacted Li^0 as a function of first cycle CE for various electrolytes. The latter was quantified by titration gas chromatography (TGC).[35,39] The solid line serves as a visual guide but does not correspond to a fit. (b) Irreversible Li^0 quantified by TGC (gray bars) and porosity quantified by focused ion beam (FIB) milling followed by computational 3D reconstruction from cryogenic transmission electron microscopy imaging (teal bars) for three representative electrolytes.[40] (c-e) Computational reconstruction of voids/pores after a plating half-cycle (1 mAh/cm²) for three representative electrolytes.[40] Teal represents voids. (b-e) were adapted from Lee *et al*[40] with permission from ACS Publications.

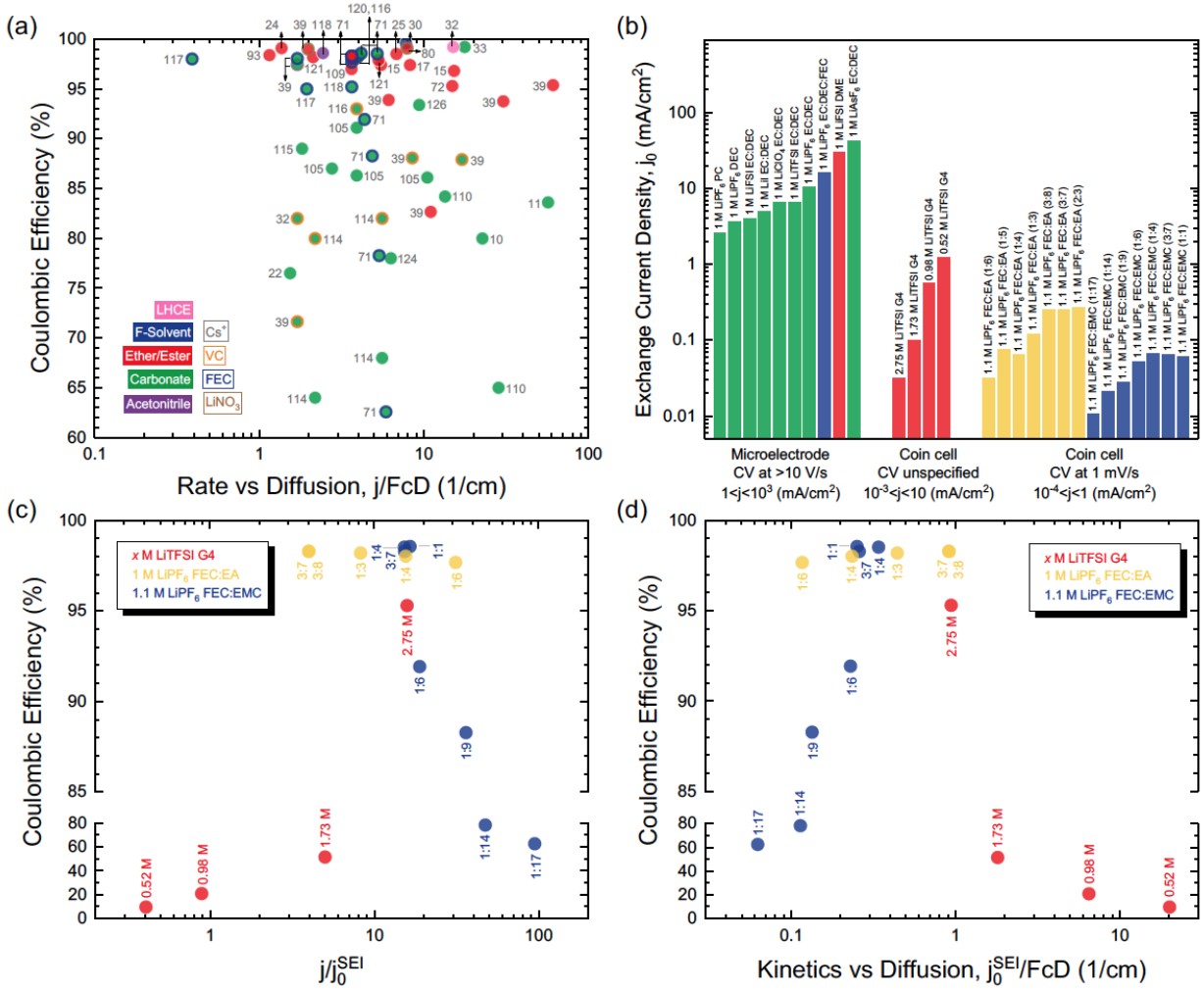

Figure 4 | Interplay between Li^+ transport and redox kinetics. (a) Influence of electrolyte Li^+ diffusivity and applied current density on Li plating and stripping CE. A detailed description of the electrolytes, their relative properties and CE is reported in the **Source Data**. (b) Exchange current density in various electrolytes, organized by measurement protocol.[50,71,72] (c) Influence of the electrolyte-dependent exchange current density j_0 on CE measured in the presence of an SEI (low scan rate, 1 mV/s).[71,72] (d) Correlation between exchange current density, Li^+ diffusivity and CE.[71,72] Calculations for Li^+ diffusivity values are shown in the **Source Data** file and references used are aggregated here.[24,28,30,32,71,73,109,118,121,125-132]